

Derivation: Qubit Encoding and the Jordan-Wigner Transform

Seminar notes

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1 Setup: The Jordan-Wigner Transform

The goal is to map the fermionic Kitaev chain to a system of qubits. Fermionic operators obey the anticommutation relations:

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \{c_i, c_j\} = 0. \quad (1)$$

Qubit operators on different sites commute: $[\sigma_i^\alpha, \sigma_j^\beta] = 0$ for $i \neq j$. To simulate fermions with qubits, we must encode the fermionic anticommutation into the qubit algebra. This is achieved using the Jordan-Wigner transform, which attaches a “string” of Pauli Z operators to each fermionic lowering operator to act as a parity counter for the preceding sites:

$$c_j = \left(\prod_{k=0}^{j-1} Z_k \right) \sigma_j^-, \quad c_j^\dagger = \left(\prod_{k=0}^{j-1} Z_k \right) \sigma_j^+. \quad (2)$$

Here, $\sigma^\pm = \frac{1}{2}(X \pm iY)$. The operators X, Y, Z are the standard Pauli matrices acting on the two-level qubit system at site j :

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3)$$

These matrices satisfy the standard Pauli algebra $X^2 = Y^2 = Z^2 = I$, anticommute with each other ($\{X, Y\} = 0$, etc.), and form the basis for observing the state of the qubit. The Z -string flips sign when an earlier site is occupied (because $Z|1\rangle = -|1\rangle$), restoring the correct anticommutation relations between operators at different sites.

2 Derivation of the Qubit Hamiltonian

We start with the Kitaev Hamiltonian in open boundary conditions:

$$H_K = -\mu \sum_{j=0}^{L-1} c_j^\dagger c_j - t \sum_{j=0}^{L-2} (c_j^\dagger c_{j+1} + \text{h.c.}) + \Delta \sum_{j=0}^{L-2} (c_j^\dagger c_{j+1}^\dagger + \text{h.c.}). \quad (4)$$

We apply the Jordan-Wigner mapping term by term.

2.1 Chemical Potential Term

The number operator at site j maps to the single-qubit Z basis:

$$c_j^\dagger c_j = \sigma_j^+ \sigma_j^- = \left(\frac{X_j + iY_j}{2} \right) \left(\frac{X_j - iY_j}{2} \right) = \frac{I - Z_j}{2}. \quad (5)$$

Summing this yields the transverse field term:

$$-\mu \sum_j c_j^\dagger c_j = -\frac{\mu}{2} \sum_j (I - Z_j) \cong \frac{\mu}{2} \sum_j Z_j \quad (\text{dropping constant offsets}). \quad (6)$$

Note: In the presentation, the sign of the transverse field is $-\frac{\mu}{2} \sum Z_j$. This arises if we map the empty fermionic state to the $|0\rangle$ state (where $Z = +1$), meaning $c_j^\dagger c_j = \frac{I - Z_j}{2}$. Dropping the $\frac{I}{2}$ gives the $-\frac{\mu}{2} Z_j$ term as written in the slides.

2.2 Hopping Term

For the hopping term $c_j^\dagger c_{j+1}$, the Z -strings partially cancel:

$$\begin{aligned} c_j^\dagger c_{j+1} &= \left(\prod_{k=0}^{j-1} Z_k \right) \sigma_j^+ \cdot \left(\prod_{m=0}^j Z_m \right) \sigma_{j+1}^- \\ &= \sigma_j^+ Z_j \sigma_{j+1}^- = \sigma_j^+ (I - 2\sigma_j^+ \sigma_j^-) \sigma_{j+1}^- \\ &= -\sigma_j^+ \sigma_{j+1}^-. \end{aligned}$$

Adding its Hermitian conjugate:

$$c_j^\dagger c_{j+1} + \text{h.c.} = -(\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+) = -\frac{X_j X_{j+1} + Y_j Y_{j+1}}{2}. \quad (7)$$

Multiplying by $-t$, we get the XX and YY coupling.

2.3 Pairing Term

Similarly, for the pairing term $c_j^\dagger c_{j+1}^\dagger$:

$$c_j^\dagger c_{j+1}^\dagger = \sigma_j^+ Z_j \sigma_{j+1}^+ = -\sigma_j^+ \sigma_{j+1}^+.$$

Adding its Hermitian conjugate:

$$c_j^\dagger c_{j+1}^\dagger + \text{h.c.} = -(\sigma_j^+ \sigma_{j+1}^+ + \sigma_j^- \sigma_{j+1}^-) = -\frac{X_j X_{j+1} - Y_j Y_{j+1}}{2}. \quad (8)$$

Multiplying by Δ , this generates an asymmetric XX and YY coupling.

2.4 Full Qubit Hamiltonian

Combining all terms, we obtain the Kitaev Hamiltonian in the qubit basis:

$$\begin{aligned} H &= -\frac{\mu}{2} \sum_j Z_j + t \sum_j \left(\frac{X_j X_{j+1} + Y_j Y_{j+1}}{2} \right) - \Delta \sum_j \left(\frac{X_j X_{j+1} - Y_j Y_{j+1}}{2} \right) \\ &= -\frac{\mu}{2} \sum_j Z_j + \frac{t - \Delta}{2} \sum_j X_j X_{j+1} + \frac{t + \Delta}{2} \sum_j Y_j Y_{j+1}. \end{aligned} \quad (9)$$

This is the transverse-field XY model.

3 The Sweet Spot $t = \Delta$

At the “sweet spot” ($t = \Delta$), the $X_j X_{j+1}$ terms perfectly cancel. The Hamiltonian simplifies to a transverse-field Ising-like model but with YY interactions:

$$H = -\frac{\mu}{2} \sum_j Z_j + t \sum_j Y_j Y_{j+1}. \quad (10)$$

If $\mu = 0$ (deep inside the topological phase), the Hamiltonian is purely $H = t \sum_j Y_j Y_{j+1}$. The boundary operators X_0 and X_{L-1} do not appear in this Hamiltonian, meaning they form zero-energy modes that are exactly decoupled from the bulk. These correspond to the localized Majorana edge modes.

4 Fermion Parity and Topological Degeneracy

4.1 Fermion Parity Operator

The total fermion parity operator is defined as the parity of the number of fermions:

$$P_f = (-1)^{\sum_j c_j^\dagger c_j}. \quad (11)$$

Using the number operator identity $c_j^\dagger c_j = (I - Z_j)/2$, each factor becomes $(-1)^{(1-Z_j)/2} = Z_j$. Thus, the total parity operator in the qubit basis is simply the product of all Z operators:

$$P = \prod_{j=0}^{L-1} Z_j. \quad (12)$$

4.2 Conservation and Parity Gap

Because the Kitaev Hamiltonian only creates or destroys fermions in pairs (via the Δ term) and moves them one by one (via the t term), the total parity is conserved: $[H, P] = 0$. The eigenstates can be block-diagonalized into the even ($P = +1$) and odd ($P = -1$) parity sectors.

In the topological phase, the emergence of Majorana zero modes provides a zero-energy excitation that toggles the parity without changing the overall energy of the system. In a finite system, the left and right Majorana modes overlap, lifting the exact degeneracy by an energy $E_0(L) \sim (|\mu|/2t)^L$. The difference in energy between the lowest even and odd states is exactly this **parity gap**:

$$|E_0^{(+)} - E_0^{(-)}| \approx 2t \left(\frac{|\mu|}{2t} \right)^L. \quad (13)$$

This gap vanishes exponentially with the system size L , manifesting the bulk-boundary correspondence in the many-body spectrum.

4.3 Many-Body Qubit Spectrum vs Finite-Size BdG Spectrum

The **finite-size BdG spectrum** describes single-particle quasiparticle excitations. Because of particle-hole symmetry, these excitation energies always come in $\pm E$ pairs. The lowest positive energy E_0 represents the energy cost to add a single Bogoliubov quasiparticle to the ground state. In the topological phase, this lowest excitation is the hybridized Majorana mode, and $E_0 \rightarrow 0$ exponentially as $L \rightarrow \infty$.

The **many-body qubit spectrum**, on the other hand, represents the total energies of all 2^L many-body states of the system. Because $[H, P] = 0$, every many-body state belongs to either the even parity sector (blue curves in the presentation) or the odd parity sector (red curves). The lowest energy state in the even sector $E_0^{(+)}$ is the overall ground state of the system with an even number of fermions. The lowest energy state in the odd sector $E_0^{(-)}$ is the ground state with an odd number of fermions.

Why Even and Odd Parity Curves Do Not Overlap (Finite-Size Effects): In an infinite chain ($L \rightarrow \infty$) inside the topological phase, the left and right Majorana edge modes are infinitely separated and exact zero-energy modes. Toggling the occupation of the non-local fermionic state formed by these two Majoranas changes the parity of the system without costing any energy. Hence, the even and odd many-body ground states become perfectly degenerate, and the curves would exactly overlap.

However, in a **finite-size** chain, the left and right Majorana modes are not perfectly localized; their wavefunctions decay exponentially but have a finite overlap in the bulk of the chain. This finite overlap causes the Majorana modes to hybridize and split away from exact zero energy by the amount $E_0(L)$.

Because toggling the parity now requires occupying this hybridized mode, it costs a small, non-zero energy $E_0(L)$. Therefore, the odd-parity ground state lies slightly higher in energy than the even-parity ground state:

$$E_0^{(-)} - E_0^{(+)} = E_0(L) \approx 2t \left(\frac{|\mu|}{2t} \right)^L. \quad (14)$$

This is why the blue (even) and red (odd) curves in the many-body spectrum do not perfectly overlap for finite L . The gap between them is precisely the finite-size BdG splitting. As one tunes μ towards the critical point $|\mu| = 2t$, the Majorana localization length (coherence length) diverges. The overlap between the edge modes becomes macroscopic, the splitting $E_0(L)$ becomes large, and the parity gap visibly opens, eventually merging into the bulk gap in the trivial phase.